

LECTURE 14: WEAK INSTRUMENTS AND MANY INSTRUMENTS

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Economics 272, GSB 507, Spring 2025

OUTLINE

1. Introduction
2. Motivation: The Angrist-Krueger Study
3. Single Weak Instrument
4. Many Weak Instruments

INTRODUCTION

- Standard normal asymptotic approximation to sampling distribution of IV, TSLS, and LIML estimators relies on **non-zero correlation** between instruments and endogenous regressors.
- If correlation is close to zero, these approximations are not accurate even in fairly large samples.
- In the just identified case
 - Not too much to worry about (Angrist & Kolesar, 2024)
 - TSLS/LIML confidence intervals will still be fairly wide in most cases, even if not valid, unless degree of endogeneity is very high.
 - If concerned with this, alternative confidence intervals are available that are valid uniformly.
 - No better estimators available.

RECOMMENDATION FOR OVER-IDENTIFIED CASE

- (I don't often give explicit recommendations)
- In the case with large degree of overidentification TSLS has poor properties: considerable bias towards OLS, and substantial underestimation of standard errors.
- LIML is much better in terms of bias, but its regular standard error is **not** correct in overidentified case.
- A simple multiplicative adjustment to conventional LIML standard errors based on Bekker asymptotics or (Bayesian) random effects likelihood (Chamberlain & Imbens) works well.
- **Overall Recommendation: use LIML, with Bekker-adjusted standard errors.**
(as long as instruments are valid, see Kolesar *et al* (2015))

MOTIVATION FOR MANY WEAK INSTRUMENT CASE: ANGRIST-KRUEGER 1991

- AK were interested in estimating the returns to years of education. Their basic specification is:

$$Y_i = \alpha + \beta \cdot E_i + \varepsilon_i,$$

where Y_i is log (yearly) earnings and E_i is years of education.

- In an attempt to address the endogeneity problem AK exploit variation in schooling levels that arise from differential impacts of compulsory schooling laws by quarter of birth and use quarter of birth as an instrument.
- This leads to IV estimate (using only 1st and 4th quarter data):

$$\hat{\beta}^{IV} = \hat{\beta}^{TSLS} = \hat{\beta}^{LIML} = \frac{\bar{Y}_4 - \bar{Y}_1}{\bar{E}_4 - \bar{E}_1} = 0.089 \text{ (s.e. 0.011)}$$

AK WITH MANY INSTRUMENTS

- AK also present estimates based on additional instruments. They take the basic 3 job dummies and interact them with 50 state and 9 year of birth dummies, for about 180 instruments.
- Here (following Chamberlain and Imbens, 2004) we interact the single binary instrument with state times year of birth dummies to get 500 instruments.
- Also including the state times year of birth dummies as exogenous covariates leads to the following model:

$$Y_i = X_i^\top \beta + \varepsilon_i, \quad \mathbb{E}[Z_i \cdot \varepsilon_i] = 0,$$

- Here X_i is the 501-dimensional vector with the 500 state/year dummies and years of education, and Z_i is the vector with 500 state/year dummies and the 500 state/year dummies multiplying the indicator for the fourth quarter of birth.

TSLS *versus* LIML

- The TSLS estimator for β is

$$\hat{\beta}_{\text{TSLS}} = 0.073 \text{ (s.e. 0.008)}$$

suggesting the extra instruments improve the standard errors a little bit.

- However, LIML estimator tells a somewhat **different** story,

$$\hat{\beta}_{\text{LIML}} = 0.095 \text{ (s.e. 0.017)}$$

with an increase in the standard error.

- Under conventional asymptotics (fixed number of instruments, fixed parameters, number of observations going to infinity)

$$\sqrt{N} \left(\hat{\beta}_{\text{LIML}} - \hat{\beta}_{\text{TSLS}} \right) \longrightarrow 0$$

- So LIML and TSLS estimates and standard errors should be close.

BOUND-JAEGER-BAKER CRITIQUE

- BJB suggest that despite the large (census) samples used by AK asymptotic normal approximations may be very poor because the instruments are only very weakly correlated with the endogenous regressor.
- The most striking evidence for this is based on the following calculation (based on suggestion by Alan Krueger).
 - Take the AK data and re-calculate their estimates after replacing the actual quarter of birth dummies by random indicators with the same marginal distribution.
 - In principle this means that the standard (gaussian) large sample approximations for TSLS and LIML are invalid since they rely on non-zero correlations between the instruments and the endogenous regressor: the ratio of normals centered at zero is Cauchy). Estimators would not be consistent.

RESULTS FROM KRUEGER SUGGESTION

	Single Instrument		500 Instruments			
	TSLS/LIML		TSLS		LIML	
Real QOB	0.088	(0.022)	0.078	(0.009)	0.111	(0.019)
bootstrap var		{0.022}		{0.008}		{1.001}
bootstrap quant		{0.023}		{0.008}		{0.110}
Bekker		[0.022]				[0.039]
Random QOB	-0.131	(0.152)	0.072	(0.010)	0.539	(0.266)
bootstrap var		{1.685}		{0.008}		{4.768}
bootstrap quant		{0.373}		{0.009}		{1.291}
Bekker		[0.152]				[3.581]

- With single random instrument conventional standard errors are not so bad (but normal approximation is not valid).
- With many random instruments the results are troubling. The conventional standard errors **erroneously** suggest that the (useless) instruments can be used to infer precisely the returns to education.

SIMULATIONS WITH A SINGLE INSTRUMENT

- 10,000 artificial data sets, all of size 160,000, designed to mimic the AK data.
- In each of these data sets half the units have quarter of birth (denoted by Q_i) equal to 0 and 1 respectively.

$$\begin{pmatrix} \nu_i \\ \eta_i \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0.446 & \rho \cdot \sqrt{0.446} \cdot \sqrt{10.071} \\ \rho \cdot \sqrt{0.446} \cdot \sqrt{10.071} & 10.071 \end{pmatrix} \right).$$

- The correlation between the reduced form residuals in the AK data is $\rho = 0.318$. Here we consider a range values for ρ .

$$Y_i = 5.892 + 0.014 \cdot Q_i + \nu_i,$$

$$E_i = 12.688 + 0.151 \cdot Q_i + \eta_i,$$

SIMULATIONS WITH A SINGLE INSTRUMENT

- Now we calculate the IV estimator and its standard error, using either the actual job variable or a random job variable as the instrument.
- We are interested in the size of tests of the null that coefficient on years of education is equal to 0.089 (is equal to the ratio = $0.014/0.151$).
- We base the test on the t-statistic. Thus we reject the null if the ratio of the point estimate minus 0.089 and the standard error is greater than 1.96 in absolute value.
- We repeat this for 12 different values of the reduced form error correlation. In the table we report the coverage rate and the median and 0.10 quantile of the width of the estimated 95% confidence intervals.

COVERAGE RATES OF CONVENTIONAL INTERVALS TSLS CONFIDENCE INTERVALS BY DEGREE OF ENDOGENEITY

ρ	0.0	0.4	0.6	0.8	0.9	0.95	0.99
implied OLS	0.00	0.08	0.13	0.17	0.19	0.20	0.21

Real QOB

Cov rate	0.95	0.95	0.96	0.95	0.95	0.95	0.95
Med Width 95% CI	0.09	0.08	0.07	0.06	0.05	0.05	0.05
0.10 quant Width	0.08	0.07	0.06	0.05	0.04	0.04	0.04

Random QOB

Cov rate	0.99	1.00	1.00	0.98	0.92	0.82	0.53
Med Width 95% CI	1.82	1.66	1.45	1.09	0.79	0.57	0.26
0.10 quant Width	0.55	0.51	0.42	0.33	0.24	0.17	0.08

COVERAGE RATES OF CONVENTIONAL TSLS CONFIDENCE INTERVALS BY DEGREE OF ENDOGENEITY

- In this example, unless the reduced form correlations are **very high**, e.g., at least 0.95, with irrelevant instruments the conventional confidence intervals are wide and have good coverage.
- The amount of endogeneity that would be required for the conventional confidence intervals to be misleading is larger than one typically encounters in cross-section settings.
- Put differently, although formally conventional confidence intervals are not valid uniformly over the parameter space (e.g., Dufour, 1997), the subsets of the parameter space where results are substantively misleading may be of limited interest.
- This in contrast to the case with many weak instruments where especially TSLS can be misleading in empirically relevant settings.

A SINGLE WEAK INSTRUMENT

- Structural form of the model

$$Y_i = \beta_0 + \beta_1 \cdot X_i + \varepsilon_i,$$

$$X_i = \pi_0 + \pi_1 \cdot Z_i + \eta_i,$$

- $(\varepsilon_i, \eta_i) \perp\!\!\!\perp Z_i$, and jointly normal with covariance matrix Σ .
- The **reduced form** for the first equation is

$$Y_i = \alpha_0 + \alpha_1 \cdot Z_i + \nu_i,$$

- The parameter of interest is $\beta_1 = \alpha_1/\pi_1$. Let

$$\Omega = \mathbb{E} \left[\begin{pmatrix} \nu_i \\ \eta_i \end{pmatrix} \cdot \begin{pmatrix} \nu_i \\ \eta_i \end{pmatrix}' \right], \quad \text{and} \quad \Sigma = \mathbb{E} \left[\begin{pmatrix} \varepsilon_i \\ \eta_i \end{pmatrix} \cdot \begin{pmatrix} \varepsilon_i \\ \eta_i \end{pmatrix}' \right]$$

A SINGLE WEAK INSTRUMENT

- Standard IV estimator:

$$\hat{\beta}_1^{IV} = \frac{\frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y}) (Z_i - \bar{Z})}{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X}) (Z_i - \bar{Z})},$$

- **Concentration parameter** (measures strength of instrument):

$$\lambda = \pi_1^2 \cdot \sum_{i=1}^N (Z_i - \bar{Z})^2 / \sigma_{\eta}^2.$$

A SINGLE WEAK INSTRUMENT

- Normal approximations for numerator and denominator are accurate:

$$\sqrt{N} \left(\frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y}) (Z_i - \bar{Z}) - \text{Cov}(Y_i, Z_i) \right) \approx \mathcal{N}(0, V(Y_i \cdot Z_i)),$$

$$\sqrt{N} \left(\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X}) (Z_i - \bar{Z}) - \text{Cov}(X_i, Z_i) \right) \approx \mathcal{N}(0, V(X_i \cdot Z_i)).$$

- If $\pi_1 \neq 0$, as the sample size gets large, then the ratio will eventually be well approximated by a normal distribution as well.
- However, if $\text{Cov}(X_i, Z_i) \approx 0$, the ratio may be better approximated by a Cauchy distribution, as the ratio of two normals centered close to zero.

STAIGER-STOCK ASYMPTOTICS AND UNIFORMITY

- Staiger and Stock investigate the distribution of the standard IV estimator under an **alternative** asymptotic approximation.
- The standard asymptotics (strong instrument asymptotics in the Staiger-Stock terminology) is based on fixed parameters and the sample size getting large.
- In their alternative asymptotic sequence Staiger-Stock model π_1 as a function of the sample size so that the concentration parameter converges to a constant:

$$\pi_{1N} = c/\sqrt{N}, \quad \lambda \longrightarrow c^2 \cdot V(Z_i).$$

- Staiger-Stock then compare coverage properties of various confidence intervals under this (weak instrument) asymptotic sequence.

STAIGER-STOCK ASYMPTOTICS AND UNIFORMITY

- The importance of the Staiger-Stock approach is in demonstrating for any sample size there are values of the nuisance parameters such that the actual coverage is substantially away from the nominal coverage.
- More recently the issue has therefore been reformulated as requiring confidence intervals to have asymptotically the correct coverage probabilities uniformly in the parameter space (for all π_1). See for a discussion from this perspective Mikusheva.
- Note that there **cannot** exist estimators that are **consistent** for β^* **uniformly** in the parameter space since if $\pi_1 = 0$, there are no consistent estimators for β_1 .
- However, for testing there are generally **confidence intervals** that are **uniformly valid**, but they are not of the conventional form, that is, a point estimate plus or minus a constant times a standard error.

ANDERSON-RUBIN CONFIDENCE INTERVALS

- Let the instrument $\dot{Z}_i = Z_i - \bar{Z}$ be measured in deviations from its mean. Then define the statistic

$$S(\beta_1) = \frac{1}{N} \sum_{i=1}^N \dot{Z}_i \cdot (Y_i - \beta_1 \cdot X_i).$$

- Then,
 - under the null hypothesis that $\beta_1 = \beta_1^*$,
 - conditional on the instruments,
 - given normality of the error terms,

the statistic $\sqrt{N} \cdot S(\beta_1^*)$ has an **exact** normal distribution

$$\sqrt{N} \cdot S(\beta_1^*) \sim \mathcal{N} \left(0, \sum_{i=1}^N \dot{Z}_i^2 \cdot \sigma_\varepsilon^2 \right).$$

ANDERSON-RUBIN CONFIDENCE INTERVALS

- Anderson and Rubin (1949) propose basing tests for the null hypothesis

$$H_0 : \beta_1 = \beta_1^0, \quad \text{against the alternative hypothesis } H_a : \beta_1 \neq \beta_1^0,$$

- Consider the statistic

$$\text{AR}(\beta_1^0) = \frac{N \cdot S(\beta_1^0)^2}{\sum_{i=1}^N \tilde{Z}_i^2} \cdot \left(\begin{pmatrix} 1 & -\beta_1^0 \end{pmatrix} \Omega \begin{pmatrix} 1 \\ -\beta_1^0 \end{pmatrix} \right)^{-1} \sim \chi^2(1)$$

A confidence interval can be based on this test statistic by inverting it:

$$\text{CI}_{0.95}^{\beta_1} = \{ \beta_1 \mid \text{AR}(\beta_1) \leq 3.84 \}$$

- This interval can be equal to the whole real line.
- Not attractive from a Bayesian perspective – corresponds to posterior probability 1, not 0.95.

ANDERSON-RUBIN WITH K INSTRUMENTS

- The reduced form is

$$X_i = \pi_0 + \pi_1' Z_i + \eta_i,$$

- $S(\beta_1^0)$ is now normally distributed vector.
- AR statistic with associated confidence interval:

$$AR(\beta_1^0) = N \cdot S(\beta_1^0)' \left(\sum_{i=1}^N \tilde{Z}_i \cdot \tilde{Z}_i' \right)^{-1} S(\beta_1^0) \cdot \left(\begin{pmatrix} 1 & -\beta_1^0 \end{pmatrix} \Omega \begin{pmatrix} 1 \\ -\beta_1^0 \end{pmatrix} \right)^{-1}.$$

$$CI_{0.95}^{\beta_1} = \left\{ \beta_1 \mid AR(\beta_1) \leq \chi_{0.95}^2(K) \right\},$$

- The problem is that this confidence interval can be **empty** because it simultaneously tests validity of instruments.
- Very unattractive from Bayes perspective – if empty it has posterior probability

KLEIBERGEN TEST

- Kleibergen modifies AR statistic through

$$\tilde{S}(\beta_1^0) = \frac{1}{N} \sum_{i=1}^N \left(\tilde{Z}_i' \hat{\pi}_1(\beta_1^0) \right) \cdot \left(y_i - \beta_1^0 \cdot x_i \right),$$

- Here $\hat{\pi}$ is the maximum likelihood estimator for π_1 under the restriction $\beta_1 = \beta_1^0$.
- The test is then based on the statistic

$$K(\beta_1^0) = \frac{N \cdot \tilde{S}(\beta_1^0)^2}{\sum_{i=1}^N \tilde{Z}_i^2} \cdot \left(\begin{pmatrix} 1 & -\beta_1^0 \end{pmatrix} \Omega \begin{pmatrix} 1 \\ -\beta_1^0 \end{pmatrix} \right)^{-1}.$$

- This has an approximate chi-squared distribution, and can be used to construct a confidence interval.

MOREIRA'S SIMILAR TESTS

- Moreira (2003) proposes a method for adjusting the critical values that applies to a number of tests, including the Kleibergen test.
- His idea is to focus on **similar** tests, test that have the same rejection probability for all values of the nuisance parameter (the π) by adjusting critical values (instead of using quantiles from the chi-squared distribution).
- The way to adjust the critical values is to consider the distribution of a statistic such as the Kleibergen statistic conditional on a complete sufficient statistic for the nuisance parameter.
- In this setting a complete sufficient statistic is readily available in the form of the maximum likelihood estimator under the null, $\hat{\pi}_1(\beta_1^0)$.

MOREIRA'S SIMILAR TESTS

- Moreira's preferred test is based on the likelihood ratio. Let

$$LR\left(\beta_1^0\right)=2 \cdot\left(L\left(\hat{\beta}_1, \hat{\pi}\right)-L\left(\beta_1^0, \hat{\pi}\left(\beta_1^0\right)\right)\right),$$

be the likelihood ratio.

- Then let $c_{LR}(p, 0.95)$, be the 0.95 quantile of the distribution of $LR\left(\beta_1^0\right)$ under the null hypothesis, conditional on $\hat{\pi}\left(\beta_1^0\right)=p$.
- The proposed test is to reject the null hypothesis at the 5% level if

$$LR\left(\beta_1^0\right)>c_{LR}\left(\hat{\pi}\left(\beta_1^0\right), 0.95\right),$$

- Here conventional test would use critical values from a chi-squared distribution with a single degree of freedom.
- The critical values are tabulated for low values of K .

This test can then be converted to construct a 95% confidence intervals.

CONDITIONING ON THE FIRST STAGE

- These confidence intervals are asymptotically valid irrespective of the strength of the first stage (the value of π_1). However, they are not valid if one first inspects the first stage, and conditional on the strength of that, decides to proceed.
- Specifically, if in practice one first inspects the first stage, and decide to abandon the project if the first stage F-statistic is less than some fixed value, and otherwise proceed by calculating confidence interval, the large sample coverage probabilities would not be the nominal ones.
- Chioda and Jansson propose a confidence interval that is valid conditional on the strength of the first stage.
- A caveat is that this involves loss of information, and thus the Chioda-Jansson confidence intervals are wider than confidence intervals that are not valid conditional on the first stage.

MANY (WEAK) INSTRUMENTS

- Here we discuss the case with many weak instruments. This makes things somewhat **easier**.
- The problem is twofold:
 - The bias in the standard estimators,
 - The misleadingly small standard errors based on conventional procedures, leading to poor coverage rates for standard confidence intervals in many situations.
- Resampling methods such as bootstrapping do not solve these problems, see earlier table.
- The literature has taken a number of approaches.
 - Part of the literature has focused on alternative confidence intervals analogues to the single instrument case.
 - In addition a variety of new point estimators have been proposed.
- Generally LIML still does well, but standard errors need to be adjusted.

BEKKER ASYMPTOTICS

- Bekker (1995) derives large sample approximations for TSLS and LIML based on sequences where the number of instruments increases proportionally to the sample size.
- He shows that TSLS is **not** consistent in that case.
- LIML **is** consistent, but the conventional LIML standard errors are **not** valid.
- Bekker then provides LIML standard errors that are valid under this asymptotic sequence. Even with relatively small numbers of instruments the differences between the Bekker and conventional asymptotics can be substantial.

BEKKER ASYMPTOTICS

- Bekker correction, single endogenous regressor:

$$Y_i = \beta'_1 X_{1i} + \beta'_2 X_{2i} + \varepsilon_i = \beta' X_i + \varepsilon_i,$$

$$X_{1i} = \pi'_1 Z_{1i} + \pi'_2 X_{2i} + \eta_i = \pi' Z_i + \eta_i.$$

- Define the matrices $\mathbf{P}_Z$ and $\mathbf{M}_Z$ as:

$$\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}', \quad \mathbf{M}_Z = \mathbf{I} - \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'.$$

- Let σ^2 be the variance of ε_i , with consistent estimator $\hat{\sigma}^2$. The standard TSLS variance is

$$\hat{\mathbf{V}}_{\text{tsls}} = \hat{\sigma}^2 \cdot (\mathbf{X}\mathbf{P}_Z\mathbf{X})^{-1}.$$

BEKKER ASYMPTOTICS

- Under the standard, fixed number of instrument asymptotics, the asymptotic variance for LIML is identical to that for TSLS, and so in principle we can use the same estimator.
- In practice researchers typically estimate the variance for LIML as

$$\hat{V}_{\text{LIML}} = \hat{\sigma}^2 \cdot \left(\mathbf{X} \mathbf{P}_Z \mathbf{X} - \hat{\lambda} \cdot \mathbf{X}' \mathbf{M}_Z \mathbf{X} \right)^{-1}.$$

BEKKER ASYMPTOTICS

- To get Bekker's correction, we need a little more notation. Define

$$\Omega = \begin{pmatrix} \mathbf{Y} & \mathbf{X} \end{pmatrix} \mathbf{P}_Z \begin{pmatrix} \mathbf{Y} & \mathbf{X} \end{pmatrix} / N = \begin{pmatrix} \Omega_{11} & \Omega_{12} \\ \Omega'_{12} & \Omega_{22} \end{pmatrix},$$

$$\Omega_{11} = \mathbf{Y} \mathbf{P}_Z \mathbf{Y} / N, \quad \Omega_{12} = \mathbf{Y} \mathbf{P}_Z \mathbf{X} / N, \quad \text{and } \Omega_{22} = \mathbf{X} \mathbf{P}_Z \mathbf{X} / N.$$

$$\mathbf{A} = N \cdot \frac{\Omega'_{12} \Omega_{12} - \Omega_{22} \beta \Omega_{12} - \Omega'_{12} \beta' \Omega_{22} + \Omega_{22} \beta \beta' \Omega_{22}}{\Omega_{11} - 2 \Omega_{12} \beta + \beta' \Omega_{22} \beta}.$$

- Then:

$$\hat{\mathbf{V}}_{\text{bekker}} = \hat{\sigma}^2 \cdot \left(\mathbf{X} \mathbf{P}_Z \mathbf{X} - \hat{\lambda} \mathbf{X}' \mathbf{M}_Z \mathbf{X} \right)^{-1} (\mathbf{X} \mathbf{P}_Z \mathbf{X} - \lambda \mathbf{A}) \left(\mathbf{X} \mathbf{P}_Z \mathbf{X} - \hat{\lambda} \mathbf{X}' \mathbf{M}_Z \mathbf{X} \right)^{-1}$$

- Recommended in practice

RANDOM EFFECTS ESTIMATORS

- Chamberlain and Imbens (2004) propose a random effects quasi maximum likelihood (REQML) estimator.
- They propose modelling the first stage coefficients π_k , for $k = 1, \dots, K$, in the regression

$$X_i = \pi_0 + \pi_1' Z_i + \eta_i = \pi_0 + \sum_{k=1}^K \pi_k \cdot Z_{ik} + \eta_i,$$

(after normalizing the instruments to have mean zero and unit variance,) as independent draws from a normal $\mathcal{N}(\mu_\pi, \sigma_\pi^2)$ distribution.

- Implicit prior for maximum likelihood is flat on the whole real line, which is not attractive for the AK study: effect of QOB on years of education should be less than 1 (year).

RANDOM EFFECTS ESTIMATORS

- Assuming also joint normality for (ε_i, η_i) , one can derive the likelihood function

$$\mathcal{L}(\beta_0, \beta_1, \pi_0, \mu_\pi, \sigma_\pi^2, \Omega).$$

- In contrast to the likelihood function in terms of the original parameters $(\beta_0, \beta_1, \pi_0, \pi_1, \Omega)$, this likelihood function depends on a small set of parameters, and a quadratic approximation to its logarithms is more likely to be accurate.

RANDOM EFFECTS ESTIMATORS

- CI discuss some connections between the REQML estimator and LIML and TSLS in the context of this parametric set up.
- First they show that in large samples, with a large number of instruments, the TSLS estimator corresponds to the restricted maximum likelihood estimator where the variance of the first stage coefficients is fixed at a large number, or $\sigma_{\pi}^2 = \infty$:

$$\hat{\beta}_{\text{tsls}} \approx \arg \max_{\beta_0, \beta_1, \pi_0, \mu_{\pi}} L(\beta_0, \beta_1, \pi_0, \mu_{\pi}, \sigma_{\pi}^2 = \infty, \Omega).$$

- From a Bayesian perspective, TSLS corresponds approximately to the posterior mode given a flat prior on all the parameters, and thus puts a large amount of prior mass on values of the parameter space where the instruments are jointly powerful.

RANDOM EFFECTS ESTIMATORS

- In the special case where we fix $\mu_\pi = 0$, and Ω is known, and the random effects specification applies to all instruments, CI show that the REQML estimator is identical to LIML.
- However, like the Bekker asymptotics, the REQML calculations suggests that the standard LIML variance is too small: the variance of the REQML estimator is approximately equal to the standard LIML variance times

$$1 + \sigma_\pi^{-2} \cdot \left(\left(\begin{pmatrix} 1 \\ \beta_1 \end{pmatrix} \right)' \Omega^{-1} \begin{pmatrix} 1 \\ \beta_1 \end{pmatrix} \right)^{-1}.$$

- This is similar to the Bekker adjustment if we replace σ_π^2 by $\sum_{i=1} (\pi_1' \tilde{Z}_i)^2 (K \cdot N)$ (keeping in mind that the instruments have been normalized to have unit variance).
- In practice the CI adjustment will be bigger than the Bekker adjustment because the ml estimator for σ_π^2 will take into account noise in the estimates of the $\hat{\pi}$, and so $\hat{\sigma}_\pi^2 < \sum_{i=1} (\hat{\pi}_1' \tilde{Z}_i)^2 (K \cdot N)$

CHOOSING THE NUMBER OF INSTRUMENTS

- Donald and Newey (2001) consider the problem of choosing a subset of an infinite sequence of instruments.
- They assume the instruments are ordered, so that the choice is the number of instruments to use.
- The criterion they focus on is based on an estimable approximation to the expected squared error. version of this leads to approximately the same expected squared error as using the infeasible criterion.
- Although in its current form not straightforward to implement, this is a very promising approach that can apply to many related problems such as generalized method of moments settings with many moments.

FLORES' SIMULATIONS

- In one of the more extensive simulation studies Flores-Lagunes reports results comparing TSLS, LIML, Fuller, Bias corrected versions of TSLS, LIML and Fuller, a Jackknife version of TSLS (Hahn, Hausman and Kuersteiner), and the REQML estimator, in settings with 100 and 500 observations, and 5 and 30 instruments for the single endogenous variable. Does not include LIML with Bekker standard errors.
- He looks at median bias, median absolute error, inter decile range, coverage rates.
- Flores-Lagunes concludes that “our evidence indicates that the random-effects quasi-maximum likelihood estimator outperforms alternative estimators in terms of median point estimates and coverage rates.”

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